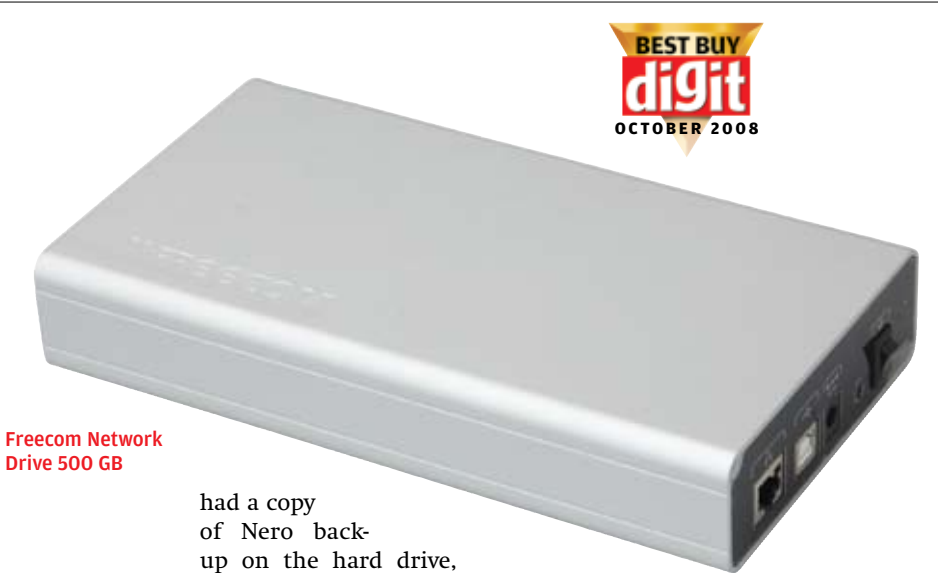


Interface

USB has been the choice of interface for all external storage devices, be it flash drives or external hard drives. And all drives tested here, except for a notable few, were equipped with full speed USB 2.0 and were backward compatible with USB 1.1. Freecom's Tough drive Pro was the only drive to come with FireWire 400 as the default port, but also had provision for USB 2.0. Western Digital's Passport Studio was another drive to come with both Firewire as well as USB 2.0. The very presence of FireWire makes these two drives Macintosh ready, and this is prominently displayed on their respective boxes. When testing, we found the FireWire devices performing better than USB 2.0, and with good reason. Though FireWire offers better performance, don't blindly opt for it unless you are using a Macintosh. Very few PCs have a FireWire port; only high-end motherboards feature them. FireWire has two standards: the older is called FireWire 400 and the newer is FireWire 800. Most drives have FireWire 400 and the ones which support FireWire 800 generally bundle an 800 to 400 adapter cable, as these two standards are not pin compatible. Sadly, none of the drives in this test came with the new eSATA interface.

Bundle

Portable drives are often bundled with software to help with setting up backup solutions or encrypting data to prevent misuse. Freecom offers NTT's Shadow as the backup tool, which is easy to configure and use. Western Digital has its proprietary tools called WD Sync and also bundles in Google Apps such as the toolbar, Picasa, Earth etc. Verbatim



Freecom Network Drive 500 GB

had a copy of Nero back-up on the hard drive, while LaCie throws in One-Click Backup and the LaCie Sync tool.

Drives such as the Western Digital Passport Studio and Freecom Tough Drive Pro come with both USB and FireWire cables, while other drives with a USB port come bundled with a USB cable. FireWire is the primary port on the Freecom drive, and it is integrated into a small pull out cable. Such an arrangement is good only if you have a laptop with FireWire. For FireWire ports at the back, an extension cable will be required, which isn't bundled. Except for LaCie, no other vendors bundled a carry case — and we think this is a must.

Capacity

While internal drives offer 1 TB capacity, portable external drives do not yet. Most portable external drives use the smaller 2.5-inch laptop drives, rather than the 3.5-inch internal hard drives. 500 GB is the highest capacity you can get in portable drives today. However, they haven't yet made it to Indian shores, as they are expensive. 320 GB is the main capacity every vendor is tout-

ing in India, while 80, 120 and 160 GB are common. Before you decide on the capacity, work out the price per GB and then make an appropriate purchase.

Observations

When testing these drives we came across some interesting observations — some good, others irritating, but they all matter. After plugging in the Verbatim drives, a small blue LED would light up, but during transfer it doesn't flicker to indicate activity. Verbatim drives were the only drives to have small slits near the sides which should help with better heat dissipation. Other drives have the hard drive sealed inside an outer shell with no air vents; their plastic bodies trap the heat — not advisable. In all three WD drives the LED indicators are placed alongside the interface ports; this is completely pointless, because you end up not seeing the indicators because the USB connectors mostly face away from you. In the case of LaCie, the blue activity indicator LED is hard to spot and the retractable USB cable is very short — good for a laptop, but it requires

Contact Sheet		External Storage Devices			
Brand	Company	Tel No	Email	Website	
Freecom	Rashi peripherals Pvt Ltd	022-67090909	response@rpotechindia.com	www.rpotechindia.com	
LaCie	neoteric infomatique ltd.	022-40859600	sales@neoteric.co.in	www.neoteric.co.in	
Verbatim	Verbatim India	NA	ramesh.tuli@verbatim.co.in	www.verbatim.com.au	
Western Digital	Western Digital	9321029204	amarjeet.singh@wdc.com	www.wdc.com	

the bundled USB extension cable when used with PCs.

Performance

Since drives were hooked on a USB port, the random access time was over 15 ms for most drives, and this affected their read/write performance in general. CPU utilisation figures were the most interesting and they showed the ill effects of the USB interface. All drives with USB, on an average, required 11 per cent of CPU time, as seen from HD Tach's CPU utilisation score. On the other hand, both the WD Studio and Freecom Tough Drive, connected to the FireWire ports, needed merely 1 to 2 per cent of CPU time. The average read / write scores were also higher in the two FireWire drives.



LaCie Rugged 500 GB

If we take the scores of synthetic benchmarks such as HD Tach and SiSoft Sandra then FireWire comes out as a better interface than USB. But if we look at the real world transfer tests, there seems to be a negligible difference in the timings. The reason being that newer generation hard drives are faster than the maximum interface speed. Both USB and FireWire are limited to approximately 400 Mbps which works out to 50 MBps (mega bytes). If we look at the SiSoft Sandra scores, all drives deliver between 35 and 38 MBps — not bad. In other words, there is a dire need for USB 3.0 to come in soon.

Among these hard drives, the Western Digital Studio is ahead of

the rest, delivering better performance figures. Trailing behind by a whisker is the Freecom Tough Drive Pro. WD's Essential 320GB delivered good figures, taking it well ahead of the other USB drives.

Conclusion

Taking features, performance and price per GB into consideration, Western Digital's Passport Studio emerges as the winner. Despite its high price per GB, the Studio makes up lost ground by offering better features and higher performance. LaCie's Little Disk 320 GB comes in second, priced sweetly at just Rs 6,000, it beats the rest of them because of its lower price per GB.

Desktop Drives

Features

Design
Western Digital's My Book series of drives is designed to appeal. Carrying forward the same theme as the Passport series, the My Book Essential has a shiny plastic shell, whereas the My Book Studio sports an anodised exterior. LED indicator lights are put to good use to enhance the visual appeal. Freecom's Network drive sports an all-around aluminium enclosure and feels solidly put together. Verbatim's External drive looks plain-jane in its black plastic enclosure. The LaCie Rugged was the smallest drive amongst these desktop

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